

Insights into Malaria Data Consistency Across Service and Disease Reporting Streams in Ethiopia's HMIS/DHIS2

Prepared by: *Enku Demssie (MACEPA Data Analytics Cohort 1 Fellow- Ethiopia)*

MACEPA Data Analytics Fellowship Program

May 2026

Abstract

Background: Malaria remains a major public health threat in Ethiopia. Effective surveillance is critical for guiding control strategies. Ethiopia's routine malaria reporting system, implemented through HMIS/DHIS2, relies on two streams, service and disease reports, each drawing from distinct primary sources. Limited evidence exists on their internal consistency. This study assessed discrepancies between these streams across temporal, geographic, and facility dimensions.

Methods: A retrospective observational analysis was conducted using national HMIS data from January 2021 to December 2025. Service reports included laboratory-confirmed malaria cases and total diagnostic tests; disease reports included species-specific confirmed cases. Data were harmonized and analyzed by reporting period, facility type, and geographic unit. Discrepancies were quantified and summarized using descriptive analyses in R version 4.5.1.

Results: During the study period, 76.9 million malaria tests were reported, yielding 25.5 million confirmed cases in service reports and 20.1 million in disease reports, with service reports exceeding disease reports by 27.3%. Discrepancies were highest at health posts (64%) and notable in private facilities (33%), while hospitals and health centers showed smaller differences. Absolute discrepancies peaked during the major transmission season, though relative differences were higher in minor and low seasons. Tigray (46%), Oromia (41%), South Ethiopia (38%), Gambella (37%), and Somali (34%) had the largest relative discrepancies. Relative differences in Somali rose after 2023, whereas Tigray declined sharply, reflecting local programmatic and contextual factors.

Conclusion: Persistent discrepancies exist between malaria service and disease reporting streams, varying by facility type, region, and season. Strengthening data management at primary health care facilities, engaging private facilities through the Public–Private Mix, conducting routine internal consistency checks, and ongoing capacity building are essential to improve malaria surveillance and support evidence-based decision-making.

Contents

Abstract	2
List of figures	3
List of tables	3
1. Introduction	4
2. Study Objective	5
3. Methods	5
4. Results	6
4.1. Overall comparison between service and disease reports	6
4.2. Magnitude of discrepancy at the national level	7
4.3. Facility type variation	7
4.4. Temporal trends (2021–2025)	8
4.5. Geographic variation	8
Discussion	9
Conclusions	10
Recommendations	11
Acknowledgement	11
References	11
Appendices	13

List of figures

Figure 1: Trends in confirmed malaria cases reported through service and disease reporting streams in Ethiopia, 2021–2025	6
Figure 2: Report selection decision tree, Source: MACEPA Data Analytics Team	13

List of tables

Table 1: Confirmed malaria cases reported through HMIS service and disease reports, Ethiopia, 2021–2025	7
--	---

1. Introduction

Malaria remains a major global public health challenge. In 2024, an estimated 282 million malaria cases were reported across 80 endemic countries, representing an increase of about 9 million cases compared with 2023 (1). The WHO African Region accounted for approximately 94% of global malaria cases, with Ethiopia among the countries contributing substantially to the regional burden. Global malaria mortality was estimated at 610,000 deaths in 2024, with Ethiopia also significantly contributing to recent increases in malaria deaths (1).

Given this substantial burden, effective surveillance systems are essential for monitoring malaria trends and guiding control strategies. Surveillance is defined as the continuous and systematic collection, analysis, interpretation, and use of disease-specific data for planning, implementing, and evaluating public health interventions (2). The Global Technical Strategy for Malaria 2016–2030 emphasizes the importance of strengthening surveillance systems and transforming malaria surveillance into a core intervention in endemic countries (3). Increasingly, routine surveillance systems are supported by digital platforms such as the District Health Information Software 2 (DHIS2), which facilitate the collection, aggregation, analysis, and use of health data for program monitoring and decision-making (2).

In Ethiopia, malaria surveillance relies primarily on the national Health Management Information System (HMIS), which is implemented through the DHIS2 platform. The system supports routine reporting of health service and disease data from health facilities and health posts and includes several malaria-related indicators used to monitor prevention, diagnosis, and treatment services (4). Ensuring the quality and consistency of these routinely reported data is critical, as reliable information is essential for effective program planning, surveillance, and evaluation (5).

Within the Ethiopian HMIS, malaria data are reported through two related reporting streams: service reports and disease reports. Disease reports capture malaria cases recorded in clinical registers, including outpatient department (OPD), inpatient department (IPD), intensive care unit (ICU), and emergency registers at health facilities, as well as disease tally sheets used at health posts. In contrast, service reports capture malaria diagnostic activities documented in laboratory registers at health facilities and service delivery tally sheets at health posts. Although both reporting streams aim to capture parasitologically confirmed malaria cases, they originate from the aforementioned different primary recording sources within the routine reporting system (6). Comparing these two reporting streams provides an opportunity to assess the internal consistency of malaria data within the routine HMIS.

Previous studies assessing routine HMIS data in Ethiopia have reported challenges related to data completeness, accuracy, and consistency across various health program indicators, including malaria (7–9). However, most of these assessments have focused on overall data quality metrics rather than examining discrepancies between specific malaria reporting streams. Consequently, limited evidence exists on the consistency between malaria service and disease reporting data within the national HMIS. Understanding these differences is important for strengthening malaria surveillance and improving the interpretation of routinely reported malaria indicators. Therefore, this study aimed to assess discrepancies between malaria service and disease reporting streams using national HMIS data in Ethiopia.

2. Study Objective

This study aimed to assess the internal consistency of malaria surveillance data within Ethiopia's routine Health Management Information System by quantifying discrepancies between confirmed malaria cases recorded in service and disease reports from January 2021 to December 2025, and by describing their temporal, geographic, and facility-type patterns.

3. Methods

A retrospective observational study was conducted to assess discrepancies between confirmed malaria cases reported through the service and disease reporting streams of Ethiopia's routine Health Management Information System (HMIS) from January 2021 to December 2025. Routine HMIS data were extracted from the national DHIS2 platform using the period, organization unit, facility type, and data element name dimensions.

Service data included the total number of malaria diagnostic tests conducted (rapid diagnostic tests and microscopy) and all laboratory-confirmed malaria positives identified through these tests without species disaggregation. Disease data included species-specific laboratory-confirmed malaria cases reported from clinical service units, excluding clinically diagnosed cases.

The extracted datasets were cleaned and harmonized by aligning reporting periods and geographic identifiers across the two reporting streams. Discrepancies between service and disease data were assessed by comparing the magnitude and direction of differences in confirmed malaria cases across reporting periods and administrative levels. Temporal, geographic, and facility-type patterns in these discrepancies were summarized using descriptive analyses. All data cleaning and analyses were conducted using R statistical software version 4.5.1.

4. Results

4.1. Overall comparison between service and disease reports

Between January 2021 and December 2025, a total of 76,919,924 parasitological malaria tests were reported through the national HMIS. During the same period, 20,054,809 laboratory-confirmed malaria cases were recorded in disease reports, while 25,532,322 confirmed malaria cases were reported through service reports. Overall, the service reporting stream recorded 5,477,513 more confirmed cases than the disease reporting stream over the five-year period, corresponding to 27.3% higher case counts.

Annual totals from both reporting streams followed a similar temporal pattern, with cases increasing steadily from 2021 to 2024 and decreasing in 2025 relative to the 2024 peak, although 2025 levels remained higher than in earlier years of the study period. Across all years, the number of confirmed malaria cases reported through the service reporting stream exceeded those recorded in disease reports (Figure 1).

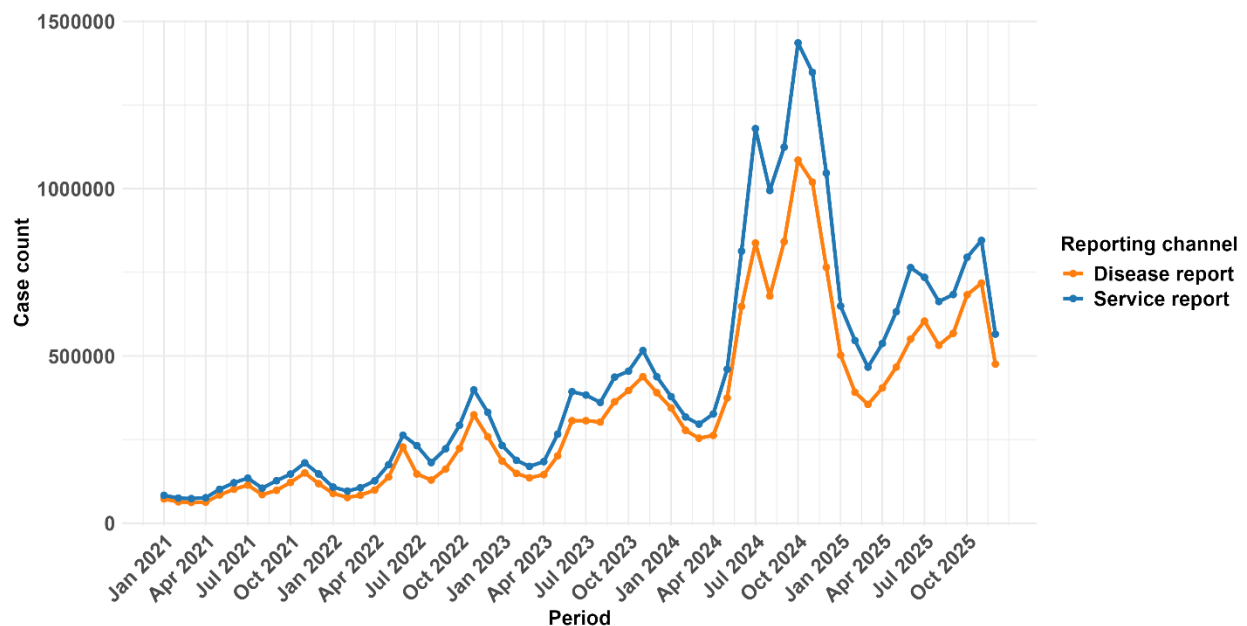


Figure 1: Trends in confirmed malaria cases reported through service and disease reporting streams in Ethiopia, 2021–2025

4.2. Magnitude of discrepancy at the national level

The magnitude of discrepancies between service and disease reporting streams varied over the study period. At the national level, the percentage difference between confirmed malaria cases was 20.8% in 2021, increasing to 29.3% in 2022, declining to 21.2% in 2023, reaching a peak of 31.6% in 2024, and decreasing to 26.1% in 2025. Across all years, service reports consistently recorded higher numbers of confirmed malaria cases than disease reports.

4.3. Facility type variation

Over the five-year study period, health centers accounted for the largest share of confirmed malaria cases, contributing 68% of cases in disease reports and 64% in service reports. Health posts followed, contributing 17% of cases in disease reports and 22% in service reports, while hospitals and private health facilities together accounted for 15% of cases in disease reports and 14% in service reports.

Examination of discrepancies between reporting streams showed variation across facility types. Health posts consistently exhibited the largest percentage difference (64%), followed by private health facilities (33%), health centers (21%), and public hospitals (11%). This pattern remained consistent across most years of the study period. The only variation occurred in 2023, when health centers recorded a slightly larger percentage discrepancy than private health facilities, while health posts and public hospitals maintained the highest and lowest discrepancies, respectively (*Table 1*).

Table 1: Confirmed malaria cases reported through HMIS service and disease reports, Ethiopia, 2021–2025

Year	Facility type	Confirmed cases from the disease report	Confirmed cases from the Service report	Absolute difference	Percentage difference
2021	Private facilities	28149	37091	8942	32%
	Health Centers	729509	829172	99663	14%
	Health Post	260529	372839	112310	43%
	Public hospitals	117699	133180	15481	13%
2022	Private facilities	46719	61202	14483	31%
	Health Centers	1268917	1581754	312837	25%
	Health Post	396766	597536	200770	51%
	Public hospitals	246947	293745	46798	19%
2023	Private facilities	115652	134217	18565	16%
	Health Centers	2343007	2819430	476423	20%
	Health Post	461271	626200	164929	36%
	Public hospitals	400989	443933	42944	11%

2024	Private facilities	176988	249737	72749	41%
	Health Centers	5119238	6206121	1086883	21%
	Health Post	1261914	2334283	1072369	85%
	Public hospitals	830524	930368	99844	12%
2025	Private facilities	263749	356425	92676	35%
	Health Centers	4164795	5006538	841743	20%
	Health Post	1083095	1737415	654320	60%
	Public hospitals	738352	781136	42784	6%

4.4. Temporal trends (2021–2025)

Months were grouped into three periods based on Ethiopia’s commonly recognized malaria transmission seasons: the minor transmission season (April–June), the major transmission season (September–December), and the low or no transmission season (January–March and July–August), in order to assess whether seasonal transmission patterns influence discrepancies between reporting streams. Over the five-year period, the largest absolute discrepancy between service and disease reports occurred during the major transmission season (2,336,076 cases), followed by the low/no transmission season (1,974,072 cases) and the minor transmission season (1,167,365 cases). In terms of percentage difference, discrepancies were highest during the minor and low transmission seasons (approximately 29%) and slightly lower during the major transmission season (approximately 25%). Discrepancies persisted across all seasonal periods, indicating consistent differences between reporting streams throughout the year.

4.5. Geographic variation

Geographic analysis revealed substantial variation in confirmed malaria cases and reporting discrepancies across regions and zones. Oromia, Amhara, South West, and South Ethiopia regions contributed the largest share of cases, accounting for 79% of confirmed cases in service reports and 77% in disease reports during the study period. Across all regions, service reports consistently exceeded disease reports, with the largest absolute differences observed in Oromia, Amhara, and South Ethiopia, followed by South West (*Table 2*).

Relative discrepancies (relative to disease report) were highest in Tigray (46%), Oromia (41%), South Ethiopia (38%), Gambella (37%), and Somali (34%), while the lowest were observed in Addis Ababa, Dire Dawa, and Afar (*Table 2*). At the zonal level, extreme discrepancies (>100%) were identified in Doolo, Nogob, Erer, and Jarar zones of the Somali region and Nuer zone of the Gambella region. In contrast, the lowest discrepancies were observed in Addis Ababa, Burji zone of South Ethiopia, Awsi/Zone 1 of Afar, Dire Dawa, and the West Guji zone of Oromia.

Yearly trends in relative discrepancies showed generally stable or declining patterns in most regions. However, Somali region exhibited a sharp increase in percentage differences after 2023, while Tigray region showed a marked decline over the study period.

Table 2: Cumulative Regional variation in reported cases in the study period

Region	Disease report	Service report	Absolute difference	Percentage difference (%)
Tigray	325169	474084	148915	45.8
Oromia	6125998	8647756	2521758	41.2
South	1961530	2711052	749522	38.2
Gambella	457551	626806	169255	37
Somali	280788	376439	95651	34.1
Central	1067323	1295913	228590	21.4
South West	2624047	3168457	544410	20.7
Sidama	624677	735786	111109	17.8
Amhara	4780074	5539078	759004	15.9
Harari	12745	14693	1948	15.3
Benishangul Gumuz	1201133	1307382	106249	8.8
Afar	493060	533410	40350	8.2
Dire Dawa	55568	56459	891	1.6
Addis Ababa	45146	45007	-139	-0.3

Discussion

This study revealed persistent discrepancies between malaria service and disease reporting streams in Ethiopia’s HMIS/DHIS2, with service reports consistently exceeding disease reports across facility types, regions, and seasons.

At the facility level, discrepancies were largest at health posts, likely reflecting the multiple responsibilities of health extension workers, limited data management training, and weaker routine data quality assessment practices. Private health facilities also showed notable differences, which may relate to the early stage of Public–Private Mix (PPM) implementation, where private providers receive limited supervision, data quality assessment, and engagement in malaria program activities.

Structural differences in data recording processes may also contribute to these discrepancies. Service reports typically originate from a single laboratory register within a facility, whereas disease reports are compiled from multiple clinical service areas such as outpatient departments,

inpatient wards, emergency units, and intensive care units. This fragmentation can increase the likelihood that some confirmed cases recorded in laboratories are not fully captured across all clinical registers used for disease reporting. Nevertheless, hospitals and health centers showed smaller discrepancies overall, likely because their more structured reporting systems, dedicated data personnel, and stronger supervision mechanisms help ensure better consolidation of information across departments.

More broadly, the findings may also reflect the still-evolving culture of routine data use for decision-making within the health system. Where awareness of the role of data in program monitoring and decision-making is limited, less attention may be given to data quality during recording, aggregation, and reporting processes. Strengthening routine data use and feedback mechanisms can therefore play an important role in improving the accuracy and consistency of surveillance data across reporting streams.

Geographically, high-burden regions, Oromia, Amhara, Southern Ethiopia, and Southwest, recorded the largest absolute discrepancies, likely reflecting how high caseloads amplify even small relative inconsistencies into substantial absolute differences. Observations during various supportive supervision and mentorship visits indicated that clinicians often prioritize patient care over recording cases in clinical registers, contributing further to underreporting in disease reports. The Somali region showed rising relative discrepancies after 2023, possibly due to limited technical capacity, skill gaps, and weak implementation of routine data quality assurance. Tigray showed a marked decline in discrepancies following a peak in 2023, likely reflecting restoration and strengthening of services after conflict-related disruptions in the preceding years.

Temporally, the largest absolute discrepancies occurred during the major transmission season, reflecting higher caseloads and staff workload, but relative differences were lower, suggesting that seasonality alone does not explain the discrepancies.

This analysis relied on aggregated HMIS/DHIS2 data and could not assess discrepancies at the individual facility level or identify specific operational causes. Variations in reporting completeness across regions and facility types may have influenced observed patterns.

Conclusions

Discrepancies between malaria service and disease reporting streams persist in Ethiopia's HMIS/DHIS2, varying by facility type, region, and season. Health posts, private facilities, and high-burden regions require targeted attention to improve reporting consistency and ensure reliable surveillance data for programmatic decision-making.

Recommendations

- Enhance data management and verification at health posts to reduce discrepancies and improve the reliability of routine malaria reporting.
- Strengthen Performance Monitoring Team (PMT) activities and conduct regular, data-focused review meetings at all levels (woreda, zonal, regional, and national) to ensure consistent reporting practices.
- Expand the Public-Private Mix (PPM) for malaria by including private facilities in supervision, data quality assurance, and malaria program activities such as trainings and review meetings.
- Provide targeted support to high-discrepancy regions, focusing on capacity building, harmonization of reporting practices, and routine oversight.
- Implement ongoing cross-stream validation (internal consistency checks) to monitor and maintain consistency between service and disease reporting streams within the HMIS.
- Promote a stronger culture of data use for decision-making at all levels, ensuring that data quality is valued during recording, aggregation, and reporting.

These measures can enhance the quality and usability of HMIS malaria data, supporting evidence-based program decisions and progress toward elimination goals.

Acknowledgement

The author acknowledges the Ethiopian Ministry of Health Malaria and Other Vector-Borne Disease Prevention and Control Desk for providing access to the national HMIS/DHIS2 data used in this analysis. I also thank the regional, zonal, district, and facility-level health workers who contribute to routine malaria data reporting within the national surveillance system, whose efforts make analyses such as this possible. I further acknowledge the PATH/MACEPA Data Analytics Team and Fellowship Program for their invaluable guidance in data analysis.

References

1. World Health Organization. World malaria report 2025: addressing the threat of antimalarial drug resistance. Geneva: WHO; 2025. Licence: CC BY-NC-SA 3.0 IGO.
2. World Health Organization. WHO guidelines for malaria. Geneva: WHO; 2025 Aug 13. Available from: <https://doi.org/10.2471/B09514> . Licence: CC BY-NC-SA 3.0 IGO.
3. World Health Organization. Global technical strategy for malaria 2016–2030, 2021 update. Geneva: WHO; 2021. Available from: <https://iris.who.int/handle/10665/342995> .

4. Ministry of Health, Ethiopia. HMIS Indicators Reference Guide. Policy, Planning, Monitoring & Evaluation Directorate; 2022.
5. World Health Organization. Improving data quality: a guide for developing countries. Geneva: WHO; 2003.
6. Ethiopia Ministry of Health. Health Management Information System (HMIS) database. Data extracted: 2022–2025. Accessed 1 December 2025.
7. Adane A, Adege TM, Ahmed MM, Anteneh HA, Ayalew ES, Berhanu D, et al. Routine health management information system data in Ethiopia: consistency, trends, and challenges. *Glob Health Action*. 2021;14(1). doi:10.1080/16549716.2020.1868961.
8. Endriyas M, Alano A, Mekonnen E, et al. Understanding performance data: health management information system data accuracy in Southern Nations, Nationalities and People's Region, Ethiopia. *BMC Health Serv Res*. 2019;19:175. doi:10.1186/s12913-019-3991-7.
9. Tilahun B, Endehabtu BF, Minyihun A, Zayede T, Nigusie A, Atnafu A, et al. Practice, enablers and barriers of health information system accountability framework in Northwest Ethiopia 2023. *BMC Med Inform Decis Mak*. 2025;25(1). doi:10.1186/s12911-025-02942-8.

Appendices

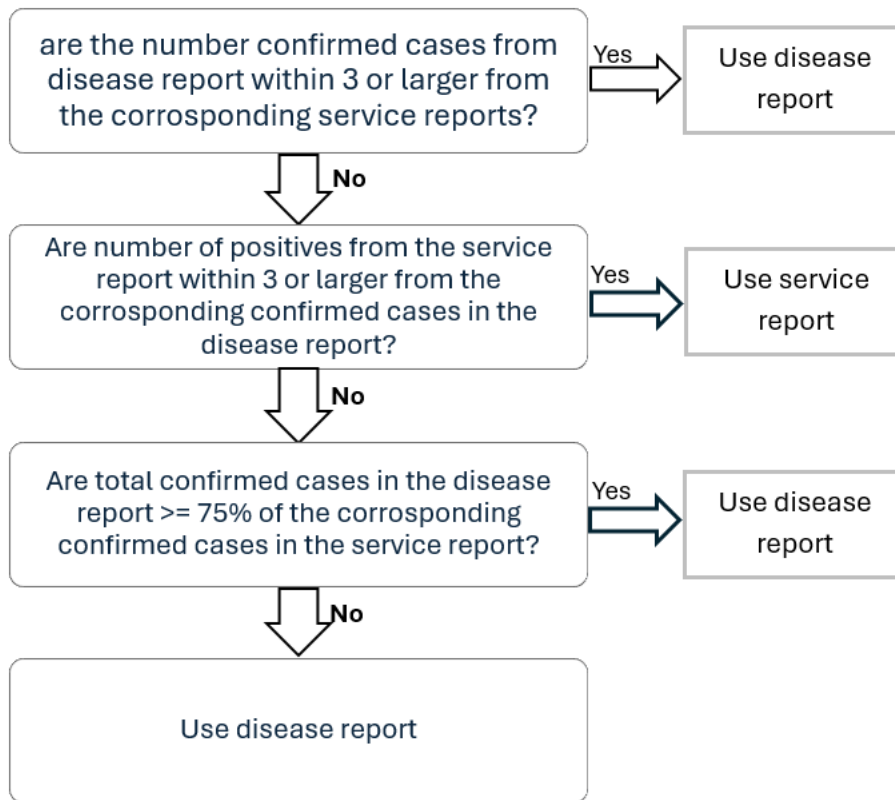


Figure 2: Report selection decision tree, Source: MACEPA Data Analytics Team